Lecture 9. Additional Topics in Probability

Poisson process, Markov chains, surprise and entropy, coding theory

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9.1 The Poisson process

Motivation. Events occurring randomly in time, with independent increments and no simultaneous events, must form a Poisson process. Counts in intervals are Poisson and the waiting times between events are exponential.

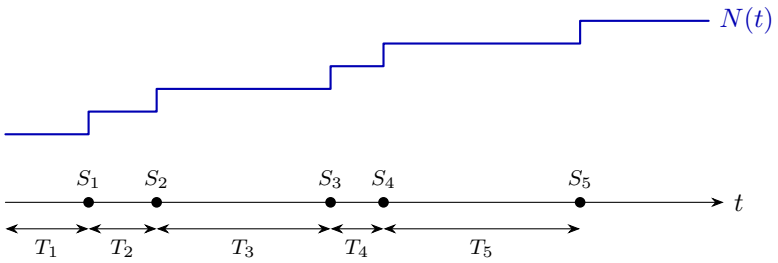


Figure. *A Poisson process: arrival times $S_1 < S_2 < \cdots$, interarrival times $T_k = S_k - S_{k-1}$ i.i.d.* Exp(λ), *and the counting process $N(t)$ (step function, one step per arrival) with $N(t) \sim$* Poisson(λt).

Definition (Poisson process)

A family $\{N(t), t \geq 0\}$ counting events in $[0, t]$ is a **Poisson process with rate** $\lambda > 0$ if (i) $N(0) = 0$;
(ii) the numbers of events in disjoint intervals are independent (independent increments);
(iii) the law of the number of events in $(s, s + t]$ depends only on t (stationary increments);
(iv) $P(N(h) = 1) = \lambda h + o(h)$ and $P(N(h) \geq 2) = o(h)$.

Proposition 9.1

For a Poisson process, $N(t) \sim$ Poisson(λt), and more generally the number of events in any interval of length t is Poisson(λt).

Proof. Split $[0, t]$ into n pieces of length t/n. By (iv) and Boole's inequality, the probability that some piece has two or more events is at most $n\, o(t/n) \to 0$. Ignoring this, $N(t)$ is the number of pieces with an event, which by (ii)–(iv) is Bin($n, \lambda t/n + o(t/n)$), and Proposition 4.4 gives the Poisson(λt) limit. ■

Let T_1 be the time of the first event and T_n the time between the $(n-1)$-th and n-th events.

Proposition 9.2

The interarrival times $T_1, T_2, \ldots$ are i.i.d. Exp(λ). Hence the time $S_n = T_1 + \cdots + T_n$ of the n-th event is Gamma(n, λ).

Proof. $P(T_1 > t) = P(N(t) = 0) = e^{-\lambda t}$. Given $T_1 = s$, the event $T_2 > t$ means no event in $(s, s + t]$, which by independent and stationary increments has probability $e^{-\lambda t}$ regardless of s; so T_2 is Exp(λ) and independent of T_1. Continue inductively. The gamma law is Proposition 6.2(a), or directly $P(S_n \leq t) = P(N(t) \geq n)$ as in Lecture 5. ■

Conversely, starting from i.i.d. exponential interarrival times produces a Poisson process; memorylessness is what makes the increments independent.

Example 9.1

Earthquakes occur as a Poisson process with rate 2 per week. The probability of at least 3 in the next 2 weeks is $1 - e^{-4}(1 + 4 + 8) = 1 - 13e^{-4} \approx 0.762$, and the probability that the next quake is more than 2 weeks away is $e^{-4} \approx 0.018$ (whatever happened before now).

Example 9.2 (Location of an event)

Given $N(t) = 1$, the time T_1 of the event is uniform on $[0, t]$: for $s \le t$,

$$P(T_1 \le s \mid N(t) = 1) = \frac{P(N(s) = 1)P(N(t) - N(s) = 0)}{P(N(t) = 1)} = \frac{\lambda s e^{-\lambda s} e^{-\lambda(t-s)}}{\lambda t e^{-\lambda t}} = \frac{s}{t}.$$

More generally, given $N(t) = n$, the event times are distributed as the order statistics of n independent $U(0, t)$ variables.

Combining with Example 6.4: if each event of a rate-λ process is independently of type 1 with probability p, the type-1 and type-2 events form *independent* Poisson processes with rates λp and $\lambda(1 - p)$.

9.2 Markov chains

Motivation. Many systems evolve with no memory beyond the current state. Markov chains model this; their n-step probabilities are matrix powers, and in the long run they often settle into a stationary distribution regardless of the start.

Definition (Markov chain)

A sequence $X_0, X_1, \ldots$ with values in $\{0, 1, \ldots, M\}$ is a **Markov chain** if $P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) = P_{ij}$ for all states and all n: the future depends on the past only through the present, with fixed **transition probabilities** $P_{ij} \ge 0$, $\sum_j P_{ij} = 1$.

The matrix $\mathbf{P} = (P_{ij})$ is the transition matrix, and

$$P(X_0 = i_0, \ldots, X_n = i_n) = P(X_0 = i_0) P_{i_0 i_1} \cdots P_{i_{n-1} i_n}.$$

Example 9.3

(a) Weather: if it rains today, it rains tomorrow with probability α; if not, with probability β. With state 0 = rain, $\mathbf{P} = \begin{pmatrix} \alpha & 1-\alpha \\ \beta & 1-\beta \end{pmatrix}$.
(b) Gambler's ruin: states $0, \ldots, N$, $P_{i,i+1} = p = 1 - P_{i,i-1}$ for $0 < i < N$, and $P_{00} = P_{NN} = 1$ (absorbing states).
(c) Ehrenfest urns: M molecules are distributed between two urns; at each step a molecule chosen at random changes urns. With X_n the number in urn 1, $P_{i,i-1} = i/M$ and $P_{i,i+1} = (M - i)/M$.
(d) If tomorrow's weather depends on the last two days, the pair (yesterday, today) is a Markov chain with four states: a process with longer memory can be made Markov by enlarging the state.

Proposition 9.3 (Chapman–Kolmogorov)

The n-step transition probabilities $P_{ij}^{(n)} = P(X_{m+n} = j \mid X_m = i)$ satisfy $P_{ij}^{(n+m)} = \sum_k P_{ik}^{(n)} P_{kj}^{(m)}$; hence $(P_{ij}^{(n)}) = \mathbf{P}^n$.

Proof. Condition on the state after n steps and use the Markov property. ∎

Example 9.4

In the weather chain with $\alpha = 0.7$, $\beta = 0.4$, the probability of rain four days after a rainy day is $(\mathbf{P}^4)_{00} = 0.5749$, and after a dry day it is $(\mathbf{P}^4)_{10} = 0.5668$: the initial state is already almost forgotten.

Theorem 9.4 (Limiting probabilities)

If for some n all entries of $\mathbf{P}^n$ are positive (the chain is *ergodic*), then $\pi_j = \lim_{n\to\infty} P_{ij}^{(n)}$ exists, does not depend on i, and is the unique solution of

$$\pi_j = \sum_i \pi_i P_{ij} \quad (j = 0, \dots, M), \qquad \sum_j \pi_j = 1.$$

The equations follow from $P_{ij}^{(n+1)} = \sum_k P_{ik}^{(n)} P_{kj}$ by letting $n \to \infty$ (existence of the limit is quoted). Thus π is a **stationary distribution**: if $X_0 \sim \pi$ then $X_n \sim \pi$ for all n. Moreover π_j is the long-run proportion of time spent in j, and $1/\pi_j$ is the mean number of steps between visits to j.

Example 9.5

(a) Weather: $\pi_0 = \alpha\pi_0 + \beta\pi_1$ and $\pi_0 + \pi_1 = 1$ give $\pi_0 = \beta/(1 + \beta - \alpha)$; for $\alpha = 0.7$, $\beta = 0.4$ this is $4/7 \approx 0.571$, matching Example 9.4.
(b) Ehrenfest: the binomial guess $\pi_j = \binom{M}{j} 2^{-M}$ satisfies $\pi_{j-1}\frac{M-j+1}{M} + \pi_{j+1}\frac{j+1}{M} = 2^{-M}\{\binom{M-1}{j-1} + \binom{M-1}{j}\} = \pi_j$, so in the long run each molecule is equally likely to be in either urn, independently. (This chain has period 2, so $P_{ij}^{(n)}$ itself does not converge, but π_j is still the long-run proportion of time in state j.)

9.3 Surprise, uncertainty and entropy

How surprised should we be to learn that an event of probability p occurred? Requiring the surprise $S(p)$ to be decreasing and continuous, with $S(1) = 0$ and $S(pq) = S(p) + S(q)$ (the surprise of two independent events adds), forces $S(p) = -C \log_2 p$; we take $C = 1$ and measure in **bits**. The functional equation is the one solved in Proposition 5.4.

The **entropy** of a discrete X with probabilities $p_1, \dots, p_n$ is the expected surprise

$$H(X) = -\sum_i p_i \log_2 p_i,$$

a measure of the uncertainty of X. It is 0 iff X is constant and at most $\log_2 n$, with equality iff X is uniform (by Jensen's inequality for the concave log: $\sum p_i \log(1/p_i) \le \log \sum p_i \cdot (1/p_i)$).

For a pair (X, Y), $H(X,Y) = -\sum_{i,j} p(x_i, y_j) \log_2 p(x_i, y_j)$, and the conditional entropy is $H_Y(X) = \sum_j H(X \mid Y = y_j) P(Y = y_j)$, the uncertainty remaining about X once Y is known. Then:

Proposition 9.5

(a) $H(X,Y) = H(Y) + H_Y(X)$.
(b) $H_Y(X) \le H(X)$, with equality iff X and Y are independent.

Proof. (a) Write $\log p(x_i, y_j) = \log p(y_j) + \log p(x_i \mid y_j)$ and take $-E$.

(b) Using $\log x \le x - 1$ (in natural logs),

$$H_Y(X) - H(X) = \sum_{i,j} p(x_i, y_j) \log_2 \frac{p(x_i)}{p(x_i \mid y_j)} \le \frac{1}{\ln 2} \sum_{i,j} p(y_j)\{p(x_i) - p(x_i \mid y_j)\} = 0,$$

with equality iff $p(x_i \mid y_j) = p(x_i)$ whenever $p(x_i, y_j) > 0$. ∎

So, on average, observing Y never increases our uncertainty about X, and $H(X) - H_Y(X)$ is the information Y carries about X.

9.4 Coding theory and entropy

We want to transmit the value of X (taking values $x_1, \dots, x_n$ with probabilities p_i) as a binary string, so that a sequence of messages can be decoded unambiguously. A code is a **prefix code** if no codeword is the beginning of another; then a concatenation is decoded by reading until a codeword is completed.

Lemma 9.6 (Kraft's inequality)

A prefix code with codeword lengths $n_1, \dots, n_N$ exists iff $\sum_i 2^{-n_i} \le 1$.

Proof. Let $L = \max n_i$. A codeword of length n_i is the prefix of 2^{L-n_i} strings of length L, and for a prefix code these sets are disjoint, so $\sum_i 2^{L-n_i} \le 2^L$. Conversely, if the inequality holds, assign codewords in order of increasing length, each time choosing a string not extending an earlier codeword; the count just given shows one is always available. ∎

Theorem 9.7 (Noiseless coding theorem)

Every prefix code for X has expected length $\sum_i p_i n_i \ge H(X)$, and there is a prefix code with expected length $< H(X) + 1$.

Proof. Let $q_i = 2^{-n_i} / \sum_k 2^{-n_k}$. By the same inequality as in Proposition 9.5(b), $-\sum p_i \log_2 p_i \le -\sum p_i \log_2 q_i = \sum p_i n_i + \log_2 \sum_k 2^{-n_k} \le \sum p_i n_i$ by Kraft. For the upper bound, take $n_i = \lceil \log_2(1/p_i) \rceil$: then $\sum 2^{-n_i} \le \sum p_i = 1$, so a prefix code exists, and $n_i < \log_2(1/p_i) + 1$. ∎

Coding blocks of k independent copies of X jointly brings the expected length per symbol within $1/k$ of $H(X)$: the entropy is the minimal number of bits per symbol needed.

Example 9.6

If X takes four values with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$, then $H(X) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{3}{8} = 1.75$ bits, and the prefix code $0, 10, 110, 111$ attains it exactly. A fixed-length code needs 2 bits.

Noisy channels. If each transmitted bit is flipped independently with probability p (binary symmetric channel), Shannon's theorem states that for every rate R below the **capacity** $C = 1 + p\log_2 p + (1-p)\log_2(1-p)$ bits per transmitted bit, there are codes whose error probability

tends to 0, while at rates above C this is impossible. Note that $C = 1 - H(\text{Bernoulli}(p))$: each received bit carries one bit minus the uncertainty introduced by the noise.

9.5 Selected exercises

Motivation. These are Ross's problems and theoretical exercises for Chapter 9 (book numbering): one computation with the Poisson process, a stationary distribution found by symmetry, and two consequences of the chain rule for entropy.

Exercise 9.1 (Two arrivals in the first hour)

Customers arrive at a bank as a Poisson process with rate λ. Given that two customers arrived during the first hour, find the probability that (a) both arrived during the first 20 minutes; (b) at least one did.

Solution.

Step 1. Measure time in hours and let $N(s)$ be the number of arrivals in $[0, s]$. By Proposition 9.1 and independent increments, $N(1/3)$ and $N(1) - N(1/3)$ are independent Poisson$(\lambda/3)$ and Poisson$(2\lambda/3)$, and $N(1) \sim$ Poisson(λ).

Step 2. For $k = 0, 1, 2$,

$$P(N(1/3) = k \mid N(1) = 2) = \frac{e^{-\lambda/3}\frac{(\lambda/3)^k}{k!}\, e^{-2\lambda/3}\frac{(2\lambda/3)^{2-k}}{(2-k)!}}{e^{-\lambda}\lambda^2/2} = \binom{2}{k}\left(\frac{1}{3}\right)^k\left(\frac{2}{3}\right)^{2-k},$$

a Bin$(2, \frac{1}{3})$ law, free of λ: each of the two arrivals behaves like an independent uniform time in the hour.

(a) $P(N(1/3) = 2 \mid N(1) = 2) = 1/9$.

(b) $P(N(1/3) \geq 1 \mid N(1) = 2) = 1 - (2/3)^2 = 5/9$. ■

Exercise 9.7 (Doubly stochastic chains)

A transition matrix on $\{0, 1, \ldots, M\}$ is doubly stochastic if $\sum_{i=0}^{M} P_{ij} = 1$ for every j. Show that if such a chain is ergodic, then $\pi_j = 1/(M+1)$ for all j.

Solution.

Step 1. Put $\pi_j = 1/(M+1)$. Then $\sum_j \pi_j = 1$ and, for every j,

$$\sum_{i=0}^{M} \pi_i P_{ij} = \frac{1}{M+1}\sum_{i=0}^{M} P_{ij} = \frac{1}{M+1} = \pi_j.$$

Step 2. By Theorem 9.4 the limiting probabilities of an ergodic chain are the unique solution of these equations, so they are the uniform probabilities. ■

Exercise 9.17 (Functions do not increase entropy)

Show that $H(f(X)) \leq H(X)$ for every discrete X and every function f.

Solution.

Step 1. Let $Y = f(X)$. The pair (X, Y) takes the value $(x, f(x))$ with probability $P(X = x)$ and every other value with probability 0, so $H(X, Y) = -\sum_x P(X = x) \log_2 P(X = x) = H(X)$.

Step 2. By Proposition 9.5(a), $H(X, Y) = H(Y) + H_Y(X)$. Each $H(X \mid Y = y)$ is the entropy of a distribution and so is nonnegative; hence $H_Y(X) \geq 0$.

Step 3. Combining, $H(Y) = H(X) - H_Y(X) \leq H(X)$, with equality iff $H_Y(X) = 0$, i.e. iff X is determined by $f(X)$ (f is one-to-one on the values X takes). ■

Exercise 9.18 (Capacity of the binary symmetric channel)

A bit X is sent and Y received, with $P(Y = 1 \mid X = 1) = P(Y = 0 \mid X = 0) = p$. The rate of transmission is $H(X) - H_Y(X)$ and the capacity is its maximum over $\alpha = P(X = 1)$. Show that the maximum is attained at $\alpha = \frac{1}{2}$ and equals $1 + p \log_2 p + (1 - p) \log_2 (1 - p)$.

Solution.

Step 1. Applying Proposition 9.5(a) in both orders, $H(X) + H_X(Y) = H(X, Y) = H(Y) + H_Y(X)$, so the rate equals $H(Y) - H_X(Y)$.

Step 2. Let $h(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$. Given $X = x$, Y equals x with probability p, so $H(Y \mid X = x) = h(p)$ for both values of x, and $H_X(Y) = h(p)$ whatever α is.

Step 3. Y takes two values, so $H(Y) \leq \log_2 2 = 1$, with equality iff $P(Y = 1) = \frac{1}{2}$. Since $P(Y = 1) = \alpha p + (1 - \alpha)(1 - p)$, the choice $\alpha = \frac{1}{2}$ gives $P(Y = 1) = \frac{1}{2}$.

Step 4. Hence the rate $H(Y) - h(p)$ is at most $1 - h(p)$, with equality at $\alpha = \frac{1}{2}$, and the capacity is $1 - h(p) = 1 + p \log_2 p + (1 - p) \log_2 (1 - p)$, the value C quoted at the end of Section 9.4. ■